

Quantum Gravitation

Problem Set 13

Yuran Seo

Due 2026-11-29 23:59

Topic

Propagation through the Cosmic Medium

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Propagation through the Cosmic Medium. State all assumptions and conventions.

$$D_T^{-1} = D_0^{-1} - \Sigma_T(\omega, k)$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$k^2(z) = k_0^2 + \Pi_R(z) + i\Pi_I(z)$$

3. Verify the third relation in two limiting regimes and explain the physical meaning of the result.

$$S_N = 2\coth(\hbar\omega/2k_B T_m) \text{Im}\Sigma_T$$

Show enough intermediate work that another student can reproduce the calculation.